

CompactionSummary[@T]

RolesFrag

Name **C** := `sys.last_compaction_turn[@T]` // 0 while no compaction has occurred

Name **lastHeadRole** :=

`sys.role_of_last_protected_head_row[@C]` // as a strict chat template counts it; none when

Name **firstTailRole** := `sys.role_of_first_surviving_tail_row[@C]`

Name **forcedUser** :=

`sys.summary_role_forced_to_user[@C]` // protected head empty (every compaction after the first

// Preferred role: the one that alternates with the head (tool rows and tool-call assistant rows are invisible to that check); an empty head or the forced case gives user. On a collision with the first tail row no standalone row is emitted and the summary is merged into that row instead (see the first replayed turn in Mark 4).

◇ If **forcedUser** or **lastHeadRole** is none or **lastHeadRole** = assistant:

◇ If **firstTailRole** ≠ user:

ROLE: USER

`sys.conversation_summary[@C]`

◆ SUMMARY_ENDS_HERE

◇ Else:

◇ If **firstTailRole** ≠ assistant:

ROLE: ASSISTANT

`sys.conversation_summary[@C]`

◆ SUMMARY_ENDS_HERE